

Customer Support Ticket Triage and Resolution using Multi-Agent AI System

Overview

Develop a multi-agent AI solution to automate the customer support ticket handling process. The system will intelligently classify incoming support requests, identify priority levels, retrieve relevant knowledge articles and historical resolutions, draft responses, and validate them against organizational compliance and communication standards before sending them to customers.

The solution aims to improve customer experience, reduce resolution times, increase agent productivity, and ensure consistent response quality.

Objectives

1. Reduce Average Resolution Time (ART).
2. Reduce ticket escalation volume.
3. Standardize customer communications.
4. Improve support team productivity.
5. Ensure compliance with company policies and regulatory requirements.

Functional Requirements

Agent 1 – Ticket Classification and Prioritization

1. Receive incoming customer support tickets.
2. Analyze ticket content and metadata.
3. Categorize tickets by issue type.
4. Determine severity and business impact.
5. Assign priority levels (Critical, High, Medium, Low).
6. Route tickets to the appropriate workflow.

Agent 2 – Knowledge Retrieval Agent

7. Search internal knowledge repositories.
8. Search historical ticket resolutions.
9. Retrieve relevant troubleshooting steps.
10. Identify similar past incidents.
11. Rank retrieved information by relevance.
12. Provide contextual information for resolution.

Agent 3 – Response Generation Agent

13. Generate draft responses using retrieved knowledge.
14. Personalize responses based on customer context.

15. Recommend resolution steps.
16. Create clear and concise communication.
17. Generate follow-up questions when information is insufficient.
18. Suggest escalation when necessary.

Agent 4 – Compliance and Quality Review Agent

19. Review drafted responses.
20. Verify adherence to company policies.
21. Check compliance with legal and regulatory requirements.
22. Validate data privacy guidelines.
23. Ensure appropriate customer communication tone.
24. Identify prohibited or risky language.
25. Approve or reject responses for delivery.

Non-Functional Requirements

26. Support near real-time processing.
27. Maintain audit logs for all agent actions.
28. Provide response traceability.
29. Support integration with ticketing systems.
30. Ensure secure handling of customer data.
31. Support scalable deployment architecture.
32. Maintain high availability and reliability.

Success Metrics

33. Reduction in average ticket resolution time.
34. Reduction in ticket escalation rate.
35. Increase in first-contact resolution rate.
36. Improvement in customer satisfaction scores.
37. Reduction in manual effort by support agents.
38. Increased consistency of customer responses.
39. Compliance accuracy above defined threshold.
40. Improved operational efficiency.

Deliverables

41. Multi-agent architecture design.
42. Agent workflow diagrams.
43. Ticket classification model.
44. Knowledge retrieval framework.
45. Response generation workflow.
46. Compliance validation framework.
47. Performance dashboard.
48. Pilot results report.

49. User documentation.

50. Final project demonstration.